

Olympiad Test Paper — Mathematics (Class 7) — Paper 3

Chapter 1: Large Numbers Around Us

Paper 3 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	1 — Large Numbers Around Us
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Watch the classic confusions — one lakh vs one million, miscounted zeros, commas grouped in the wrong system, rounding the wrong way, and place-value vs face-value slips — but here every question needs two or more clean steps, not single recall. Do not write on this paper — mark answers on your answer sheet.

1. A number has 4 in the lakhs place, 7 in the ten-thousands place, and 2 in the thousands place; every other place is 0. What is the number?

(A) 4,07,200

(B) 47,200

(C) 4,27,000

(D) 4,72,000

2. Start from 0 and press only a +1000 button. After enough presses you reach exactly 4,08,000. How many presses were made?

(A) 48

(B) 408

(C) 4080

(D) 4,08,000

3. Convert 8,05,40,000 (Indian) into the International comma system.

(A) 8,054,0000

(B) 80,540,000

(C) 805,40,000

(D) 80,54,0000

4. Round 7,46,892 to the nearest thousand, then to the nearest ten-thousand. The two results are:

(A) 7,47,000 and 7,50,000

(B) 7,46,000 and 7,40,000

(C) 7,47,000 and 7,40,000

(D) 7,46,000 and 7,50,000

- 5.** In the number 6,34,52,718, what is the sum of the place values of the two digits that are 3 and 4?
- (A) 7 (B) 30,40,000
(C) 34,00,000 (D) 3,40,000
- 6.** Which list is in strictly increasing order? 8 lakh, 0.9 million, 70,000, 1.2 million.
- (A) $70,000 < 0.9 \text{ million} < 8 \text{ lakh} < 1.2 \text{ million}$
(B) $8 \text{ lakh} < 70,000 < 0.9 \text{ million} < 1.2 \text{ million}$
(C) $70,000 < 8 \text{ lakh} < 0.9 \text{ million} < 1.2 \text{ million}$
(D) $70,000 < 8 \text{ lakh} < 1.2 \text{ million} < 0.9 \text{ million}$
- 7.** A book of stamps holds 25 stamps per page and has 4,000 pages. The total number of stamps is:
- (A) 10,000 (B) 10,00,000
(C) 1,00,000 (D) 1,00,00,000
- 8.** Estu reads '3 crore 5 lakh 60 thousand' and writes 3,56,000. What did he do wrong, and what is correct?
- (A) Nothing wrong; 3,56,000 is correct (B) He should write 30,56,000
(C) He should write 3,50,60,000
(D) He dropped the empty crore and lakh places; correct is 3,05,60,000
- 9.** Subtract: $5,00,00,000 - 2,75,000$. The answer is:
- (A) 4,97,75,000 (B) 4,97,25,000
(C) 2,25,00,000 (D) 4,72,25,000
- 10.** The smallest 8-digit number and the largest 7-digit number differ by:
- (A) 10 (B) 1
(C) 0 (D) 1,00,00,000
- 11.** Which single number lies between 9,80,000 and 10,20,000 and rounds to 10,00,000 at the nearest lakh?
- (A) 9,40,000 (B) 10,60,000
(C) 9,49,000 (D) 9,51,000
- 12.** Compute $6,000 \times 7,000$ without long multiplication. The product is:
- (A) 42,00,000 (B) 4,20,000
(C) 4,20,00,000 (D) 42,00,00,000
- 13.** In 9,87,65,432, by how much is the place value of the 8 greater than the place value of the 6?
- (A) 79,40,000 (B) 2
(C) 80,06,000 (D) 7,94,000

14. A cinema chain sold 2,45,680 tickets. Rounded to the nearest hundred and to the nearest thousand, the two figures are:

- | | |
|---------------------------|---------------------------|
| (A) 2,45,600 and 2,45,000 | (B) 2,45,700 and 2,46,000 |
| (C) 2,45,700 and 2,45,000 | (D) 2,45,600 and 2,46,000 |

15. Which comparison is TRUE?

- | | |
|--------------------------|--------------------------|
| (A) 1 lakh > 1 million | (B) 50 lakh > 6 million |
| (C) 9 crore < 80 million | (D) 3 crore > 25 million |

16. The number 'two crore three lakh nine thousand five' written in figures is:

- | | |
|-----------------|-----------------|
| (A) 2,30,90,005 | (B) 2,03,90,005 |
| (C) 2,03,09,005 | (D) 2,03,09,500 |

17. Expanded form $5 \times 10,00,000 + 2 \times 1,00,000 + 8 \times 1,000 + 4 \times 10$ represents which number?

- | | |
|---------------|---------------|
| (A) 5,28,040 | (B) 52,80,040 |
| (C) 52,08,400 | (D) 52,08,040 |

18. A water plant supplies 1,25,000 litres each day. About how much does it supply in 8 days?

- | | |
|------------------------|---------------------|
| (A) 10,00,000 litres | (B) 1,00,000 litres |
| (C) 1,00,00,000 litres | (D) 8,25,000 litres |

19. How many lakhs make 5 crore?

- | | |
|-----------|--------|
| (A) 500 | (B) 50 |
| (C) 5,000 | (D) 5 |

20. Reading right to left, the Indian system groups digits as 3, then 2, then 2, Where does the FIRST comma fall in 12345678?

- | | |
|----------------|-----------------|
| (A) 12,345,678 | (B) 1234,5678 |
| (C) 1,2345,678 | (D) 1,23,45,678 |

21. One million is what fraction of one crore?

- | | |
|---------------------|----------------------|
| (A) $\frac{1}{100}$ | (B) $\frac{1}{1000}$ |
| (C) $\frac{1}{10}$ | (D) 10 |

22. Round 8,49,99,500 to the nearest lakh.

- | | |
|-----------------|-----------------|
| (A) 8,49,00,000 | (B) 8,50,00,000 |
| (C) 9,00,00,000 | (D) 8,49,99,000 |

23. A stadium holds 60,000 fans. A tournament has 50 matches, each sold out. The total attendance is:

- | | |
|-----------------|---------------|
| (A) 3,00,000 | (B) 30,00,000 |
| (C) 3,00,00,000 | (D) 30,000 |

24. In the International system, 305,407,000 read in words is:

- (A) three hundred five million four hundred seven thousand
- (B) thirty crore five lakh forty thousand seven hundred
- (C) three hundred fifty million forty-seven thousand
- (D) thirty-five million four hundred seven thousand

25. Which of these numbers is the largest?

- (A) 95 lakh
- (B) 9 million
- (C) 99,00,000
- (D) 1 crore

26. A factory packs 144 pencils per box and ships 2,500 boxes. The total number of pencils is:

- (A) 36,000
- (B) 3,60,000
- (C) 36,00,000
- (D) 3,60,00,000

27. A warehouse holds 99,99,994 boxes. After 9 more arrive, how many boxes are there, and how many digits does the new total have?

- (A) 1,00,00,003 — 8 digits
- (B) 1,00,00,003 — 7 digits
- (C) 99,99,903 — 7 digits
- (D) 1,00,00,003 — 9 digits

28. Compute 48×125 mentally using $125 = 1000 \div 8$.

- (A) 600
- (B) 6,000
- (C) 60,000
- (D) 384

29. Arrange in DECREASING order: 7 lakh, 0.6 million, 8,00,000, 1 million.

- (A) 8,00,000 > 1 million > 7 lakh > 0.6 million
- (B) 1 million > 8,00,000 > 0.6 million > 7 lakh
- (C) 1 million > 8,00,000 > 7 lakh > 0.6 million
- (D) 7 lakh > 8,00,000 > 0.6 million > 1 million

30. The place value of a certain digit in a number is 70,00,000 and its face value is 7. In which place does the digit sit?

- (A) lakhs
- (B) crores
- (C) ten-lakhs
- (D) ten-thousands

31. A donor gave 1 crore 25 lakh rupees split equally among 25 schools. Each school received:

- (A) 5,00,000
- (B) 50,000
- (C) 50,00,000
- (D) 5,00,00,000

32. Which number, when its digits are read in the International system, equals 'four hundred seven million sixty thousand'?

- (A) 40,70,60,000
- (B) 407,060,00
- (C) 470,060,000
- (D) 407,060,000

- 33.** Estimate $4,18,000 + 5,93,000$ by rounding each to the nearest lakh first, then state the rounded sum.
- (A) $4,00,000 + 6,00,000 = 10,00,000$ (B) $4,00,000 + 5,00,000 = 9,00,000$
 (C) $5,00,000 + 6,00,000 = 11,00,000$ (D) $4,00,000 + 6,00,000 = 11,00,000$
- 34.** Compare 4,05,070 and 4,50,007. Which statement is correct?
- (A) They are equal (B) 4,50,007 is larger by 44,937
 (C) 4,05,070 is larger by 44,937 (D) 4,50,007 is larger by 45,063
- 35.** A printing press runs 12,500 copies an hour for 8 hours. Rounded to the nearest lakh, the day's output is:
- (A) 10,00,000 (B) 1,00,000
 (C) 0 (D) 2,00,000
- 36.** The digit 0 appears in how many places of the number 50,07,003?
- (A) 3 (B) 5
 (C) 4 (D) 2
- 37.** A satellite is 35,786 km above Earth. Rounded to the nearest thousand it is reported as:
- (A) 35,000 km (B) 40,000 km
 (C) 35,800 km (D) 36,000 km
- 38.** Which is the correct expanded form of 6,40,508?
- (A) $6 \times 10,00,000 + 4 \times 1,00,000 + 5 \times 100 + 8$
 (B) $6 \times 1,00,000 + 4 \times 10,000 + 0 \times 1,000 + 5 \times 100 + 0 \times 10 + 8 \times 1$
 (C) $6 \times 1,00,000 + 4 \times 10,000 + 5 \times 1,000 + 8$
 (D) $6 \times 1,00,000 + 4 \times 1,000 + 5 \times 100 + 8$
- 39.** How many ten-thousands are there in 7,00,000?
- (A) 7 (B) 700
 (C) 70 (D) 7,000
- 40.** A relief fund collected ₹3,47,82,000. Rounded to the nearest crore, the headline figure is:
- (A) 3,00,00,000 (B) 4,00,00,000
 (C) 3,50,00,000 (D) 3,48,00,000
- 41.** Mumbai's population is about 1 crore 24 lakh. Delhi's is about 1 crore 67 lakh. About how much larger is Delhi's population?
- (A) 43 thousand (B) 2 crore 91 lakh
 (C) 43 lakh (D) 57 lakh
- 42.** How many digits does the product $5,00,000 \times 4,000$ have?

(A) 10

(B) 9

(C) 11

(D) 8

43. Exactly one of these numbers rounds to 80,00,000 at the nearest ten-lakh. Which one?

(A) 76,40,000

(B) 74,00,000

(C) 85,10,000

(D) 70,90,000

44. Comparing the same physical count, which pair names the SAME amount?

(A) one lakh and one million

(B) ten lakh and one million

(C) one crore and one million

(D) ten lakh and ten million

45. A school sells 365 tickets a day for a 30-day fair. To be sure of never running short, the most sensible number of tickets to print is:

(A) round up to 11,000

(B) round down to 10,000

(C) exactly 10,950

(D) round down to 10,900

46. What is the value of $9 \times 10,00,000 + 9 \times 10,000 + 9 \times 100 + 9$?

(A) 99,99,999

(B) 90,90,099

(C) 90,90,909

(D) 9,09,909

47. Counting at one number per second, about how long does it take to count to one crore?

(A) a little under 12 days

(B) about 1,000 days

(C) about 3 years

(D) a little under 116 days

48. Which number is exactly halfway between 12,00,000 and 14,00,000?

(A) 12,50,000

(B) 13,50,000

(C) 26,00,000

(D) 13,00,000

49. In 4,08,76,005, the digit with the largest place value and the digit with face value 5 are:

(A) 4 (place value 4,00,00,000) and 5 (face value 5)

(B) 8 (place value 8,00,000) and 5

(C) 4 (place value 4,00,000) and 5

(D) 5 (place value 5,00,00,000) and 4

50. A village had 18,650 people. After a survey corrected it upward by 1,375, the new figure rounded to the nearest thousand is:

(A) 19,000

(B) 20,000

(C) 21,000

(D) 20,025

51. How many crores are there in 600 million?

(A) 6

(B) 600

(C) 6,000

(D) 60

52. Three numbers are 9,40,000, 9,04,000 and 9,00,400. Their correct increasing order is:

- (A) $9,00,400 < 9,04,000 < 9,40,000$ (B) $9,04,000 < 9,00,400 < 9,40,000$
(C) $9,40,000 < 9,04,000 < 9,00,400$ (D) $9,00,400 < 9,40,000 < 9,04,000$

53. A printing error wrote 7,30,000 as 73,000. By what factor is the wrong number too small?

- (A) 10 times too small (B) 100 times too small
(C) 7,257 less (D) 3 times too small

54. Estimate $6,12,000 \times 3$ and give the exact product. The exact product is:

- (A) 18,06,000 (B) 1,83,600
(C) 18,36,000 (D) 1,83,60,000

55. The same number is 12,030,405 in the International system. In the Indian system it is:

- (A) 12,03,04,05 (B) 1,20,30,40,5
(C) 1,20,30,405 (D) 120,30,405

56. Which comparison is correct?

- (A) $80,00,000 < 8$ lakh (B) 1 crore < 90 lakh
(C) 5 million > 5 crore (D) $6,05,000 < 60,50,000$

57. A water tank holds 12,00,000 litres. If 8,75,000 litres are used, then 3,40,000 litres are added, the litres now in the tank are:

- (A) 7,65,000 (B) 6,65,000
(C) 5,85,000 (D) 16,15,000

58. A report rounds 1,49,000 to the nearest lakh and calls it 'about 1 lakh', while another rounds 1,51,000 the same way. Which is true?

- (A) both round to 1,00,000 (B) both round to 2,00,000
(C) $1,49,000 \rightarrow 2,00,000$ and $1,51,000 \rightarrow 1,00,000$
(D) $1,49,000 \rightarrow 1,00,000$ and $1,51,000 \rightarrow 2,00,000$

59. A printing house produces 2,40,000 newspapers a day. To the nearest lakh, the output over 6 days is:

- (A) 10,00,000 (B) 1,40,000
(C) 14,00,000 (D) 15,00,000

60. In 7,06,03,040 the digit 7 and the digit 3 have place values whose difference is:

- (A) 4 (B) 7,03,000
(C) 6,96,97,000 (D) 6,99,97,000

Solutions — Mathematics (Class 7), Chapter 1: Large Numbers Around Us — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	D	21	C	31	A	41	C	51	D
2	B	12	C	22	B	32	D	42	A	52	A
3	B	13	A	23	B	33	A	43	A	53	A
4	A	14	B	24	A	34	B	44	B	54	C
5	C	15	D	25	D	35	B	45	A	55	C
6	C	16	C	26	B	36	C	46	C	56	D
7	C	17	D	27	A	37	D	47	D	57	B
8	D	18	A	28	B	38	B	48	D	58	D
9	B	19	A	29	C	39	C	49	A	59	C
10	B	20	D	30	C	40	A	50	B	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Lakhs place value = $4 \times 1,00,000 = 4,00,000$. Ten-thousands = $7 \times 10,000 = 70,000$. Thousands = $2 \times 1,000 = 2,000$. Sum = $4,00,000 + 70,000 + 2,000 = 4,72,000$. Reading 4 lakh, 0 ten-thousand, 7 thousand gives 4,07,200 — that swaps the 7 and the empty ten-thousands slot. Dropping a zero and writing 47,200 makes it a 5-digit number, losing the lakhs digit. Writing 4,27,000 puts the 2 in the ten-thousands and 7 in the thousands — the two digits are exchanged.

2. (B) Each press adds 1,000, so presses = $4,08,000 \div 1,000 = 408$. Reading the number as 'forty-eight thousand' and dividing gives 48, which only reaches 48,000. Multiplying instead of placing the comma right gives 4080, which reaches 40,80,000. The number 4,08,000 itself counts +1 presses, not +1000 presses.

3. (B) The digits are 80540000 — eight crore five lakh forty thousand. International grouping is 3-3-3 from the right: 80,540,000 (eighty million, five hundred forty thousand). Grouping as 8,054,0000 leaves a 4-digit final block, which the 3-3-3 rule

forbids. Keeping the Indian 2-2 groups on the left (805,40,000) mixes the two systems. Writing 80,54,0000 again leaves a 4-digit block at the end.

4. (A) Nearest thousand: look at the hundreds digit of 7,46,892 — it is 8 (≥ 5), so round the thousands up: $7,46,892 \rightarrow 7,47,000$. Nearest ten-thousand: look at the thousands digit of 7,46,892 — it is 6 (≥ 5), so round the ten-thousands up: $7,46,892 \rightarrow 7,50,000$. Rounding both down (7,46,000 and 7,40,000) ignores the 8 and the 6. The pair 7,47,000 and 7,40,000 rounds the first correctly but the second wrongly down. The pair 7,46,000 and 7,50,000 rounds the first wrongly down.

5. (C) Digit 3 sits in the ten-lakhs place: place value = $3 \times 10,00,000 = 30,00,000$. Digit 4 sits in the lakhs place: place value = $4 \times 1,00,000 = 4,00,000$. Sum = $30,00,000 + 4,00,000 = 34,00,000$. Adding the face values $3 + 4$ gives 7 — that ignores place. Writing 30,40,000 misplaces the 4 one position too low (ten-thousands). Writing 3,40,000 drops one zero from each place value.

6. (C) Convert all to plain numbers: 8 lakh = 8,00,000; 0.9 million = 9,00,000; 70,000 = 70,000; 1.2 million = 12,00,000. Order: $70,000 < 8,00,000 < 9,00,000 < 12,00,000$. So $70,000 < 8 \text{ lakh} < 0.9 \text{ million} < 1.2 \text{ million}$. Placing 0.9 million (9,00,000) before 8 lakh (8,00,000) reverses those two. Placing 8 lakh before 70,000 ignores that 70,000 is smallest. Ending with 0.9 million after 1.2 million reverses the largest pair.

7. (C) Total = $25 \times 4,000$. Use $25 \times 4 = 100$, then attach the three zeros of 4,000: $100 \times 1,000 = 1,00,000$ — exactly one lakh. Getting 10,000 forgets a factor of 10 in the multiplication. Getting 10,00,000 (ten lakh) attaches one zero too many. Getting 1,00,00,000 (one crore) attaches two extra zeros.

8. (D) Three crore = 3,00,00,000; five lakh = 5,00,000; sixty thousand = 60,000. Sum = $3,00,00,000 + 5,00,000 + 60,000 = 3,05,60,000$. Estu's 3,56,000 (three lakh fifty-six thousand) collapsed the crore and the lakh-zero into the thousands — far too small. Writing 30,56,000 reads it as 'thirty lakh fifty-six thousand', still missing the crore scale. Writing 3,50,60,000 reads the 5 lakh as 50 lakh.

9. (B) Five crore is 5,00,00,000. Subtract 2,75,000: $5,00,00,000 - 2,75,000 = 4,97,25,000$. Check by adding back: $4,97,25,000 + 2,75,000 = 5,00,00,000$. Getting 4,97,75,000 subtracts only 2,25,000 by mishandling the borrow. Getting 2,25,00,000 wrongly subtracts 2,75,00,000 (a lakh too large). Getting 4,72,25,000 borrows from the wrong place, dropping 25 lakh.

10. (B) Smallest 8-digit number = 1,00,00,000 (one crore). Largest 7-digit number = 99,99,999. Their difference = $1,00,00,000 - 99,99,999 = 1$. Consecutive whole numbers always differ by 1. Answering 10 confuses this with a place jump. Answering 0 wrongly treats them as equal. Answering 1,00,00,000 just repeats the smaller boundary instead of subtracting.

11. (D) Rounding to the nearest lakh keeps the lakhs digit and looks at the ten-thousands digit. A number rounds to 10,00,000 when it is in [9,50,000 , 10,49,999]. Among the options, 9,51,000 lies in that band (its ten-thousands digit 5 rounds up). It also sits between 9,80,000? No — re-read: the band that rounds to 10,00,000 is from 9,50,000 to 10,49,999, and 9,51,000 is inside it. 9,40,000 rounds to 9,00,000 (ten-thousands digit 4). 10,60,000 rounds to 11,00,000 (digit 6). 9,49,000 rounds to 9,00,000 (digit 4).

12. (C) $6 \times 7 = 42$. Count the zeros: 6,000 has 3 and 7,000 has 3, so 6 zeros in all. 42 followed by 6 zeros = 42000000 = 4,20,00,000 (four crore twenty lakh). Writing 42,00,000 uses only 5 zeros. Writing 4,20,000 uses only 4 zeros. Writing 42,00,00,000 uses 7 zeros.

13. (A) Digit 8 is in the ten-lakhs place: $8 \times 10,00,000 = 80,00,000$. Digit 6 is in the ten-thousands place: $6 \times 10,000 = 60,000$. Difference = $80,00,000 - 60,000 = 79,40,000$. Subtracting face values $8 - 6 = 2$ ignores place. Writing 80,06,000 simply joins the two place values instead of subtracting. Writing 7,94,000 drops a zero from the answer.

14. (B) Nearest hundred: tens digit is 8 (≥ 5), so round the hundreds up — 2,45,680 $\rightarrow$ 2,45,700. Nearest thousand: hundreds digit is 6 (≥ 5), so round the thousands up — 2,45,680 $\rightarrow$ 2,46,000. Rounding both down (2,45,600 and 2,45,000) ignores the 8 and the 6. The pair 2,45,700 and 2,45,000 rounds the first up but the second wrongly down. The pair 2,45,600 and 2,46,000 rounds the first wrongly down.

15. (D) Convert to plain numbers: 3 crore = 3,00,00,000 and 25 million = 2,50,00,000, so 3 crore > 25 million is true. 1 lakh = 1,00,000 while 1 million = 10,00,000, so 1 lakh is the smaller, not larger. 50 lakh = 50,00,000 while 6 million = 60,00,000, so 50 lakh is smaller. 9 crore = 9,00,00,000 while 80 million = 8,00,00,000, so 9 crore is larger, making '9 crore < 80 million' false.

16. (C) Two crore = 2,00,00,000; three lakh = 3,00,000; nine thousand = 9,000; five = 5. Sum = 2,03,09,005. Writing 2,30,90,005 reads three lakh as thirty lakh and nine thousand as ninety thousand. Writing 2,03,90,005 reads nine thousand as ninety thousand. Writing 2,03,09,500 puts the 5 in the hundreds place (five hundred) instead of the ones.

17. (D) $5 \times 10,00,000 = 50,00,000$; $2 \times 1,00,000 = 2,00,000$; $8 \times 1,000 = 8,000$; $4 \times 10 = 40$. The hundred-thousands and ten-thousands and hundreds and ones are 0. Sum = $50,00,000 + 2,00,000 + 8,000 + 40 = 52,08,040$. Writing 5,28,040 collapses the 5 ten-lakhs into the lakhs place, losing a place. Writing 52,80,040 reads 8 thousand as 8 ten-thousand. Writing 52,08,400 reads 4 tens as 4 hundreds.

18. (A) Total = $1,25,000 \times 8$. Since $125 \times 8 = 1,000$, attach the three trailing zeros: $1,000 \times 1,000 = 10,00,000$ litres (ten lakh). Getting 1,00,000 multiplies 125×8 but drops the

thousands zeros. Getting 1,00,00,000 (one crore) attaches one zero too many. Getting 8,25,000 adds 8 lakh to the daily figure instead of multiplying.

19. (A) One crore = 100 lakh. So 5 crore = $5 \times 100 = 500$ lakh. Check: $500 \text{ lakh} = 500 \times 1,00,000 = 5,00,00,000 = 5$ crore. Answering 50 uses 1 crore = 10 lakh (wrong). Answering 5,000 uses 1 crore = 1,000 lakh. Answering 5 treats crore and lakh as the same scale.

20. (D) Indian grouping from the right: a group of 3 (678), then groups of 2 (45, 23, 1): 1,23,45,678. The first comma (rightmost) separates the last three digits 678 from 45 — i.e. after the thousands. Writing 12,345,678 uses International 3-3-3 grouping. Writing 1234,5678 uses 4-4 grouping, which neither system allows. Writing 1,2345,678 mixes a 4-digit middle block in.

21. (C) One million = 10,00,000 and one crore = 1,00,00,000. Fraction = $10,00,000 \div 1,00,00,000 = 1/10$. Check: 10 million = 1 crore, so one million is a tenth of a crore. Answering $1/100$ treats one million as one lakh. Answering $1/1000$ is off by a further factor of 10. Answering 10 inverts the ratio.

22. (B) Nearest lakh keeps everything down to the lakhs digit and looks at the ten-thousands digit. In 8,49,99,500 the ten-thousands digit is 9 (≥ 5), so round the lakhs up: 8,49,xx,xxx becomes 8,50,00,000. Rounding down to 8,49,00,000 ignores the 9. Jumping to 9,00,00,000 rounds to the nearest crore instead. Writing 8,49,99,000 rounds only to the nearest thousand.

23. (B) Total = $60,000 \times 50$. Since $6 \times 5 = 30$, and there are $4 + 1 = 5$ trailing zeros (four in 60,000, one in 50, but count: 60,000 has 4 zeros, 50 has 1 zero $\rightarrow$ 5 zeros), 30 followed by 5 zeros = 30,00,000 (thirty lakh). Writing 3,00,000 (three lakh) uses one fewer zero. Writing 3,00,00,000 (three crore) uses one extra zero. Writing 30,000 ignores the matches entirely.

24. (A) International 3-3-3 reads 305 | 407 | 000 as three hundred five million, four hundred seven thousand, zero — i.e. three hundred five million four hundred seven thousand. Reading it with Indian words (crore/lakh) is the wrong system and also miscounts the digits. 'Three hundred fifty million forty-seven thousand' is 350,047,000, a different number. 'Thirty-five million four hundred seven thousand' is 35,407,000, missing a digit.

25. (D) Convert: 1 crore = 1,00,00,000; 95 lakh = 95,00,000; 9 million = 90,00,000; 99,00,000 = 99 lakh. Comparing: $1,00,00,000 > 99,00,000 > 95,00,000 > 90,00,000$. So 1 crore is the largest. 95 lakh and 99,00,000 are both under a crore. 9 million equals 90 lakh, the smallest here.

26. (B) Total = $144 \times 2,500$. Compute $144 \times 25 = 3,600$, then attach the two trailing zeros of 2,500: $3,600 \times 100 = 3,60,000$ (three lakh sixty thousand). Writing 36,000 uses one

fewer zero. Writing 36,00,000 uses one extra zero. Writing 3,60,00,000 uses two extra zeros.

27. (A) $99,99,994 + 9 = 1,00,00,003$. The count crosses from the largest 7-digit number (99,99,999) into the 8-digit range. Counting the digits of 1,00,00,003: 1-0-0-0-0-0-0-3 = 8 digits (one crore three). Saying 7 digits forgets that adding past 99,99,999 pushes into 8 digits. The figure 99,99,903 mis-adds (subtracting instead of carrying). Saying 9 digits over-counts by one place.

28. (B) $48 \times 125 = 48 \times 1000 \div 8 = 48,000 \div 8 = 6,000$. Check: $6,000 \div 125 = 48$. Getting 600 divides 48,000 by 80 instead of 8. Getting 60,000 forgets to divide by 8 and just multiplies 48×1000 then keeps a zero. Getting 384 computes 48×8 instead.

29. (C) Convert: 7 lakh = 7,00,000; 0.6 million = 6,00,000; 8,00,000 = 8,00,000; 1 million = 10,00,000. Decreasing: $10,00,000 > 8,00,000 > 7,00,000 > 6,00,000$, i.e. 1 million > 8,00,000 > 7 lakh > 0.6 million. Placing 8,00,000 above 1 million reverses the top pair. Placing 0.6 million above 7 lakh reverses the bottom pair. The fully reversed list runs smallest to largest, not largest to smallest.

30. (C) Place value $\div$ face value = $70,00,000 \div 7 = 10,00,000$, which is the value of one unit in the ten-lakhs place. So the digit is in the ten-lakhs place. The lakhs place would give place value 7,00,000. The crores place would give 7,00,00,000. The ten-thousands place would give 70,000.

31. (A) 1 crore 25 lakh = 1,25,00,000. Divide by 25: $1,25,00,000 \div 25 = 5,00,000$ (five lakh) per school. Check: $5,00,000 \times 25 = 1,25,00,000$. Getting 50,000 divides by 250. Getting 50,00,000 divides by only 2.5 (misplaced decimal). Getting 5,00,00,000 multiplies instead of dividing.

32. (D) Four hundred seven million = 407,000,000; sixty thousand = 60,000. Sum = 407,060,000, grouped 3-3-3 as 407,060,000. Writing 40,70,60,000 uses Indian commas (and is forty crore seventy lakh sixty thousand — a different read). Writing 407,060,00 leaves a 2-digit final block, illegal in 3-3-3. Writing 470,060,000 reads it as four hundred seventy million.

33. (A) Round 4,18,000 to the nearest lakh: ten-thousands digit 1 (<5), so 4,00,000. Round 5,93,000: ten-thousands digit 9 (≥ 5), so 6,00,000. Sum of rounded values = $4,00,000 + 6,00,000 = 10,00,000$. Rounding 5,93,000 down to 5,00,000 ignores the 9. Rounding 4,18,000 up to 5,00,000 ignores that $1 < 5$. The last option rounds both correctly but then adds wrongly ($4 + 6 = 10$ lakh, not 11).

34. (B) Line up by place: 4,50,007 has 5 in the ten-thousands place, while 4,05,070 has 0 there, so 4,50,007 is larger. Difference = $4,50,007 - 4,05,070 = 44,937$. Check: $4,05,070 + 44,937 = 4,50,007$. They are clearly not equal (digits differ). 4,05,070 is the smaller, not larger. The difference 45,063 comes from a subtraction borrow error.

- 35. (B)** Output = $12,500 \times 8 = 1,00,000$ exactly (since $125 \times 8 = 1,000$, then $\times 100 = 1,00,000$). Rounded to the nearest lakh, 1,00,000 stays 1,00,000. Writing 10,00,000 adds a zero. Rounding to 0 would only happen below 50,000. Writing 2,00,000 doubles the true output.
- 36. (C)** Write the 7 digits: 5-0-0-7-0-0-3. The zeros are in the ten-lakhs? No — the leading digit is 5 (ten-lakhs), then 0 (lakhs), 0 (ten-thousands), 7 (thousands), 0 (hundreds), 0 (tens), 3 (ones). Count the zeros: positions 2, 3, 5, 6 — that is 4 zeros. Counting 3 misses one. Counting 5 includes a non-zero. Counting 2 misses two.
- 37. (D)** Nearest thousand: look at the hundreds digit of 35,786 — it is 7 (≥ 5), so round the thousands up: $35,786 \rightarrow 36,000$ km. Rounding down to 35,000 ignores the 7. Rounding to 40,000 is the nearest ten-thousand, not thousand. Rounding to 35,800 is the nearest hundred.
- 38. (B)** 6,40,508 has 6 lakhs, 4 ten-thousands, 0 thousands, 5 hundreds, 0 tens, 8 ones: $6 \times 1,00,000 + 4 \times 10,000 + 5 \times 100 + 8 \times 1$. The option that writes $6 \times 10,00,000$ treats 6 as ten-lakhs, making the number ten times too big. The option with $5 \times 1,000$ misreads the 5 as thousands. The option with $4 \times 1,000$ misreads the 4 as thousands instead of ten-thousands.
- 39. (C)** $7,00,000 \div 10,000 = 70$. Check: $70 \times 10,000 = 7,00,000$. Answering 7 divides by 1,00,000 (counts lakhs). Answering 700 divides by 1,000. Answering 7,000 divides by 100.
- 40. (A)** Nearest crore: look at the ten-lakhs digit of 3,47,82,000 — it is 4 (< 5), so the crores digit stays and everything below drops to 0: 3,00,00,000. Rounding up to 4 crore ignores that $4 < 5$. The figure 3,50,00,000 rounds to the nearest fifty lakh, not crore. The figure 3,48,00,000 rounds to the nearest lakh.
- 41. (C)** Delhi – Mumbai = $1,67,00,000 - 1,24,00,000 = 43,00,000 = 43$ lakh. Answering 43 thousand drops two zeros. Answering 2 crore 91 lakh adds the two populations instead of subtracting. Answering 57 lakh subtracts 24 from 81 or similar mis-borrow — $67 - 24 = 43$, not 57.
- 42. (A)** $5,00,000 \times 4,000$: multiply $5 \times 4 = 20$, then attach zeros. 5,00,000 has 5 zeros, 4,000 has 3 zeros, total 8 zeros, plus the '20' contributes its own. 20 followed by 8 zeros = 2000000000 = 2 followed by 9 zeros = a 10-digit number ($2,00,00,00,000 = 200$ crore). Count: 2-0-0-0-0-0-0-0-0-0 = 10 digits. Answering 9 drops the carry zero from $5 \times 4 = 20$. Answering 11 adds one zero too many. Answering 8 only counts the trailing zeros.
- 43. (A)** Rounding to the nearest ten-lakh (10,00,000) sends a number to 80,00,000 when it lies in [75,00,000 , 84,99,999]. 76,40,000 is inside this band, so it rounds to 80,00,000. 74,00,000 is below 75,00,000, so it rounds down to 70,00,000. 85,10,000 is above

84,99,999, so it rounds up to 90,00,000. 70,90,000 is well below 75,00,000, so it rounds to 70,00,000.

44. (B) Ten lakh = 10,00,000 and one million = 1,000,000 = 10,00,000 — the same amount. One lakh (1,00,000) is only a tenth of one million. One crore (1,00,00,000) is ten million, not one million. Ten lakh (10,00,000) is not ten million (1,00,00,000) — they differ by a factor of 10.

45. (A) Exact need = $365 \times 30 = 10,950$. To never run short, round UP past the exact figure to a clean number: 11,000. Rounding down to 10,000 leaves 950 short. Printing exactly 10,950 leaves no margin for error and is harder to order in clean lots. Rounding down to 10,900 is still 50 short.

46. (C) $9 \times 10,00,000 = 90,00,000$; $9 \times 10,000 = 90,000$; $9 \times 100 = 900$; $9 \times 1 = 9$. The lakhs, thousands, and tens places are 0. Sum = $90,00,000 + 90,000 + 900 + 9 = 90,90,909$. Writing 99,99,999 wrongly fills every place with 9. Writing 90,90,099 misplaces the hundreds and tens. Writing 9,09,909 drops the ten-lakhs scale by one place.

47. (D) One crore seconds $\div$ 86,400 seconds per day = $1,00,00,000 \div 86,400 \approx 115.7$ days, i.e. a little under 116 days. Getting ~ 12 days uses ten lakh (one million) seconds instead of one crore. Getting 1,000 days divides by about 10,000 instead of 86,400. About 3 years ($\sim 1,095$ days) overestimates by roughly ten times.

48. (D) Midpoint = $(12,00,000 + 14,00,000) \div 2 = 26,00,000 \div 2 = 13,00,000$. Check: 13,00,000 is 1,00,000 above 12,00,000 and 1,00,000 below 14,00,000. 12,50,000 is the midpoint of 12 lakh and 13 lakh. 13,50,000 is the midpoint of 13 lakh and 14 lakh. 26,00,000 is the sum, not the midpoint.

49. (A) Leftmost digit 4 is in the crores place: place value $4 \times 1,00,00,000 = 4,00,00,000$ — the largest place value in the number. The digit with face value 5 is the final 5 (ones). The option giving 8 the largest place value is wrong because 8 (8,00,000) sits in the lakhs place, smaller than 4 crore. The option giving 4 a place value of only 4,00,000 drops two zeros. The last option swaps the roles of 4 and 5.

50. (B) Corrected count = $18,650 + 1,375 = 20,025$. Nearest thousand: hundreds digit is 0 (< 5), so round down to 20,000. Rounding to 19,000 uses the old figure 18,650 (rounds to 19,000) and forgets the correction. Rounding to 21,000 rounds the wrong way. 20,025 is the exact corrected count, not the rounded one.

51. (D) 10 million = 1 crore, so 600 million = $600 \div 10 = 60$ crore. Check: 60 crore = 60,00,00,000 and 600 million = 600,000,000 = 60,00,00,000. Answering 6 uses 100 million = 1 crore. Answering 600 treats million and crore as equal. Answering 6,000 is off by a further factor of 10.

52. (A) All share 9 lakh; compare lower places. $9,00,400 =$ nine lakh four hundred; $9,04,000 =$ nine lakh four thousand; $9,40,000 =$ nine lakh forty thousand. Increasing: $9,00,400 < 9,04,000 < 9,40,000$. Placing $9,04,000$ below $9,00,400$ swaps the smallest two. The decreasing order reverses the whole list. Putting $9,40,000$ before $9,04,000$ mid-list misorders the larger two.

53. (A) $7,30,000 \div 73,000 = 10$, so the misprint is one-tenth of the true value — 10 times too small (one comma group dropped). It is not 100 times (that would be 7,300). The difference $7,30,000 - 73,000 = 6,57,000$, not 7,257. It is not 3 times — the leading digits 73 are unchanged; only the scale shifted by one place group.

54. (C) $6,12,000 \times 3 = 18,36,000$ (since $612 \times 3 = 1,836$, then attach the three zeros). Writing 18,06,000 multiplies 602×3 by mishandling the middle digit. Writing 1,83,600 drops a zero. Writing 1,83,60,000 adds a zero.

55. (C) The digits are 12030405 (8 digits) = one crore twenty lakh thirty thousand four hundred five. Indian 3-2-2-2 from the right: 405, 30, 20, 1 $\rightarrow$ 1,20,30,405. Writing 12,03,04,05 leaves a 2-digit final block (Indian system also takes a 3-digit final block). Writing 1,20,30,40,5 leaves a single trailing digit. Writing 120,30,405 mixes International and Indian grouping.

56. (D) 6,05,000 (six lakh five thousand) is far smaller than 60,50,000 (sixty lakh fifty thousand), so $6,05,000 < 60,50,000$ is true. 80,00,000 (eighty lakh) is much greater than 8 lakh, so '80,00,000 < 8 lakh' is false. 1 crore (1,00,00,000) is greater than 90 lakh (90,00,000), so '1 crore < 90 lakh' is false. 5 million = 50,00,000 is far less than 5 crore = 5,00,00,000, so '5 million > 5 crore' is false.

57. (B) Start 12,00,000. Use 8,75,000: $12,00,000 - 8,75,000 = 3,25,000$. Add 3,40,000: $3,25,000 + 3,40,000 = 6,65,000$ litres. Getting 7,65,000 mis-subtracts ($12,00,000 - 8,75,000$ as 4,25,000). Getting 5,85,000 adds only 2,60,000 in the second step. Getting 16,15,000 adds both amounts instead of subtracting the used water.

58. (D) Nearest lakh looks at the ten-thousands digit. For 1,49,000 it is 4 (<5) $\rightarrow$ 1,00,000. For 1,51,000 it is 5 (≥ 5) $\rightarrow$ 2,00,000. So they round to different lakhs. Saying both go to 1,00,000 ignores the 5 in 1,51,000. Saying both go to 2,00,000 ignores the 4 in 1,49,000. The last option rounds each the wrong way.

59. (C) Output = $2,40,000 \times 6 = 14,40,000$. Nearest lakh: the ten-thousands digit is 4 (<5), so round down to 14,00,000. Rounding to 10,00,000 forgets the multiplication is 6 days, not roughly 4. Writing 1,40,000 drops a zero from the product. Rounding to 15,00,000 rounds the wrong way ($4 < 5$).

60. (D) Digit 7 is in the crores place: $7 \times 1,00,00,000 = 7,00,00,000$. Digit 3 is in the thousands place: $3 \times 1,000 = 3,000$. Difference = $7,00,00,000 - 3,000 = 6,99,97,000$.

Subtracting face values $7 - 3 = 4$ ignores place. Writing 7,03,000 simply joins the two place values. Writing 6,96,97,000 mis-borrows, dropping 3 lakh too many.